

DESIGN AND SIMULATION OF PLASMA-INTERACTION IN HYBRID ROCKET COMBUSTION

A major project report

submitted in partial fulfilment of the requirements for the award of the degree

BACHELOR OF TECHNOLOGY
in
MECHANICAL ENGINEERING

by

K. SOMANATH SAI **22211A0324**

CH. BERITH JESHURUN 22211A0361

Under the guidance of

Dr. A. Padma Rao
Professor, Mechanical Engineering



Department of Mechanical Engineering

B. V. RAJU INSTITUTE OF TECHNOLOGY (AUTONOMOUS)

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Vishnupur, Narsapur, Medak District - 502 313

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CH. BERITH JESHURUN 22211A0361

(Dr. A. Varun)
Head of the Department

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DECLARATION

We hereby declare that the entire major project work embodied in this dissertation entitled “**DESIGN AND SIMULATION OF PLASMA-INTERACTION IN HYBRID ROCKET COMBUSTION**” has been independently carried out by us. As per my knowledge, no part of this work has been submitted for any degree in any institution, university, and organization previously.

Name of the student

Signature

K. SOMANATH SAI

CH. BERITH JESHURUN

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K. SOMANATH SAI

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SDG MAPPING

Project Title:	DESIGN AND SIMULATION OF PLASMA-INTERACTION IN HYBRID ROCKET COMBUSTION
Type of Project:	Analysis
Multidisciplinary Project (Yes/No)	Yes

Directions:

A - Look through the list of SDG in the table below and pick the SDG that most aligns with your module content. Click on any of the goals to launch a detailed description in your browser which includes specific SDG Targets.

B - Enter a score between 0 to 5 in the blue shaded area of the table below where:

0 = no coverage - unrelated SDG

1 = indirectly or unknowingly including topics related to the SDG

2 = provides insight into the SDG from the perspective of one's own discipline

3 = uses inter- or trans-disciplinary approach to address at least one SDG Target (specific sub-goal)

4 = assessment of SDG knowledge or skills in relation to course and/or program learning outcomes

5 = Students act to address the SDGs with community or industry partners

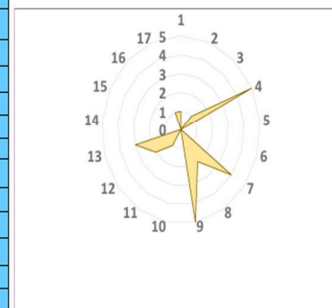
C - View your spiky radar graphs below. The first graph displays all 17 SDGs while the second displays your top score in the three SDG pillars.

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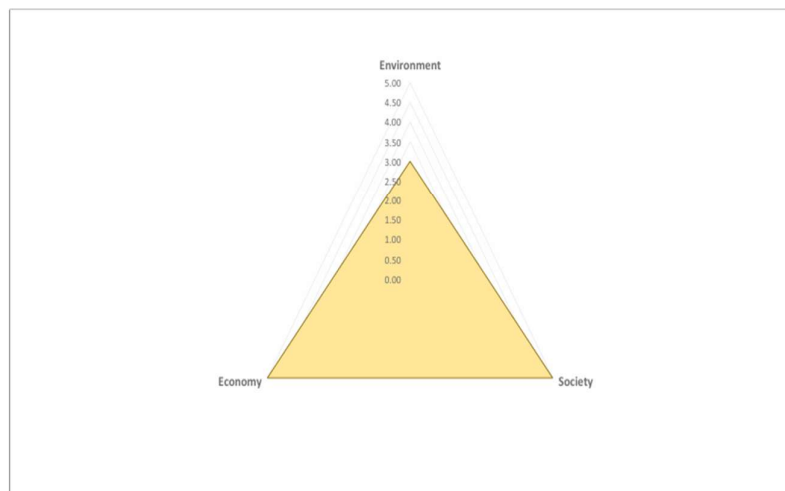
Individual SDGs		SDG Pillar	Score (0-5)
1	No Poverty	Social	1
2	Zero Hunger	Social	0
3	Good Health and Well-being	Social	1
4	Quality Education	Social	5
5	Gender Equality	Social	0
6	Clean Water and Sanitation	Environment	0
7	Affordable and Clean Energy	Social	4
8	Decent Work and Economic Growth	Economy	2
9	Industry, Innovation and Infrastructure	Economy	5
10	Reduced Inequality	Economy	0
11	Sustainable Cities and Communities	Social	1
12	Responsible Consumption and Production	Economy	2
13	Climate Action	Environment	3
14	Life Below Water	Environment	0
15	Life on Land	Environment	0
16	Peace & Justice Strong Institutions	Social	0
17	Partnerships to achieve SDGs	Social	1

VIEW RESULTS IN THE GRAPH BELOW



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The SDG Pillars	High Score (0-5)
Environment	3.00
Society	5.00
Economy	5.00



ABSTRACT

Plasma-assisted propulsion has progressed substantially since the early demonstration of pulsed plasma thrusters in the 1960s for spacecraft attitude control. Subsequent developments, such as Hall-effect thrusters, have established plasma-based systems as reliable and efficient solutions for satellite orbit raising and station-keeping.

More advanced concepts, including the Variable Specific Impulse Magneto plasma Rocket (VASIMR), helicon radio-frequency thrusters, and microwave-based electrodeless plasma devices, have been explored to achieve higher thrust levels, extended operational lifetimes, and variable specific impulse for future deep-space missions. These developments highlight the increasing importance of plasma technologies in modern aerospace propulsion research.

Building upon these advancements, this project investigates a laboratory-scale hybrid rocket thrust chamber incorporating plasma-assisted combustion, where plasma is used exclusively as a combustion enhancement mechanism rather than as a direct thrust-producing medium.

The study establishes a realistic design-level framework for integrating plasma-assisted combustion into hybrid rocket propulsion systems, providing a foundation for future experimental validation and high-fidelity plasma–combustion coupling studies.

Objectives

- To design a laboratory-scale hybrid rocket thrust chamber and converging–diverging nozzle suitable for plasma-assisted combustion studies.
- To develop a numerical simulation framework using analytical methods and CFD to analyze internal flow behaviour and thrust generation.
- To conceptually integrate nanosecond pulsed plasma energy deposition within the oxidizer–fuel mixing region to enhance combustion characteristics.
- To analyze key flow parameters such as pressure, temperature, velocity, and Mach number distribution within the combustion chamber and nozzle.
- To evaluate the feasibility and potential advantages of plasma-assisted combustion in hybrid rocket propulsion systems.

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NOMENCLATURE

Abbreviations

CFD	Computational Fluid Dynamics
RPA	Rocket Propulsion Analysis
HV	High Voltage
GOX	Gaseous Oxygen
NSPD	Nanosecond Pulsed Discharge
BVRIT	B V Raju institute of technology

Symbols

O ₂	Oxygen
A	Cross-section area (m ²)
CH ₂	Hydrocarbon fuel
L*	Characteristic chamber length
A _t	Nozzle throat area (m ²)
A _e	Nozzle exit area (m ²)
A _c	Combustion chamber area (m ²)
A _e /A _t	Nozzle expansion ratio
D _c	Combustion chamber diameter (mm)
D _t	Nozzle throat diameter (mm)
D _e	Nozzle exit diameter (mm)
L _c	Combustion chamber length (mm)
L _e	Nozzle length (mm)
F	Thrust force (N)

$\dot{m}$	Mass flow rate (kg/s)
V_e	Exhaust velocity (m/s)
P_e	Nozzle exit pressure (Pa)
P_c	Combustion chamber pressure (pa)
P_a	Ambient pressure (Pa)
T	Temperature (K)
T_c	Combustion chamber temperature (K)
T_e	Nozzle exit temperature (K)
M	Mach number
R	Gas content kJ/(kg·K)
C_p	Specific heat at constant pressure kJ/(kg·K)
C^*	Characteristic velocity (m/s)
I_{sp}	Specific impulse (s)
O/F	Oxidizer-to-fuel ratio

Greek letters

γ	Ratio of specific heats
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CHAPTER I

INTRODUCTION

1.1 Background of Rocket Propulsion

Rocket propulsion is one of the most fundamental technologies enabling space exploration, satellite deployment, and high-speed aerospace applications. The principle of rocket propulsion is based on Newton's third law of motion, which states that for every action there is an equal and opposite reaction. In a rocket engine, high-temperature and high-pressure combustion gases are accelerated through a nozzle and expelled at very high velocity. The momentum of the exhaust gases produces a reaction force that propels the vehicle in the opposite direction.

Unlike conventional propulsion systems that rely on atmospheric oxygen, rockets carry both fuel and oxidizer, allowing them to operate in space where atmospheric oxygen is unavailable. Chemical rockets remain the most widely used propulsion systems due to their high thrust generation capability and relatively simple operational mechanisms. These propulsion systems convert chemical energy stored in propellants into thermal energy through combustion, which is subsequently converted into kinetic energy of the exhaust gases.

The performance of a rocket engine largely depends on the efficiency of combustion and the expansion of gases through the nozzle. Therefore, understanding the interaction between combustion processes, thermodynamic properties of propellants, and compressible flow dynamics is essential for designing efficient propulsion systems.

1.2 Classification of Rocket Propulsion Systems

Rocket propulsion systems are generally classified into three main categories based on the physical state of the propellants used: solid rocket propulsion, liquid rocket propulsion, and hybrid rocket propulsion.

Solid rocket engines contain both fuel and oxidizer mixed together in a solid propellant grain. These engines are mechanically simple, highly reliable, and capable of producing large thrust levels. However, they lack controllability because once ignited, the combustion process continues until all the propellant is consumed.

Liquid rocket engines store fuel and oxidizer separately in liquid form and mix them in the combustion chamber during operation. Liquid propulsion systems offer excellent controllability, including throttle capability and engine restart. Despite these advantages, liquid rocket engines are mechanically complex and require sophisticated feed systems, pumps, and cryogenic storage.

Hybrid rocket propulsion combines features of both solid and liquid propulsion systems. In hybrid rockets, the fuel is stored in solid form while the oxidizer is supplied in liquid or gaseous form. The oxidizer flows through the combustion chamber and reacts with the solid fuel surface, generating combustion gases that are accelerated through a nozzle to produce thrust. Hybrid rockets provide several advantages such as improved safety, simpler design compared to liquid rockets, and better controllability compared to solid rockets.

1.3 Fundamentals of Hybrid Rocket Combustion

As shown in Fig. 1.1. combustion in a hybrid rocket engine occurs at the interface between the solid fuel grain and the injected oxidizer flow. The oxidizer enters the combustion chamber through an injector and flows over the fuel surface, where heat from combustion causes the solid fuel to vaporize or decompose. The resulting gaseous fuel then reacts with the oxidizer to produce high-temperature combustion products.

The combustion process is governed by a boundary layer diffusion flame, where mixing and reaction occur near the fuel surface. The rate at which the fuel surface recedes, known as the fuel regression rate, is a key parameter influencing thrust generation and overall engine performance.

Common fuels used in hybrid rockets include HTPB and paraffin-based fuels, with paraffin showing higher regression rates due to enhanced mass transfer characteristics. However, challenges such as limited regression rate, mixing inefficiencies, and ignition reliability continue to affect hybrid rocket performance.

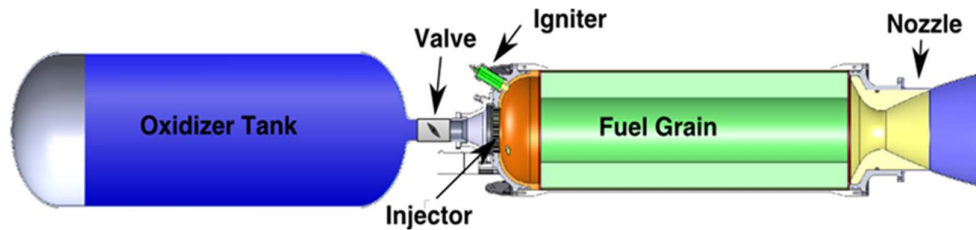


Fig. 1.1. Illustration of a Hybrid Rocket Propulsion

1.4 Converging–Diverging Rocket Nozzle

The expansion of combustion gases in a rocket engine occurs through a converging–diverging nozzle, commonly known as the De Laval nozzle. The nozzle is designed to accelerate high-pressure combustion gases from subsonic velocities within the combustion chamber to supersonic velocities at the exit.

In the converging section of the nozzle, the flow accelerates as the cross-sectional area decreases. At the throat, which represents the minimum area of the nozzle, the flow reaches Mach 1, known as the choked flow condition. Beyond the throat, in the diverging section, the flow continues to accelerate and becomes supersonic as the nozzle area increases.

The performance of the nozzle is governed by compressible flow relations such as the area–Mach number relation, pressure ratio conditions, and thermodynamic properties of the combustion gases. Proper nozzle design is essential for maximizing thrust generation and ensuring efficient conversion of thermal energy into kinetic energy.

1.5 Challenges in Hybrid Rocket Propulsion

Despite their advantages, hybrid rocket engines face several technical challenges that limit their widespread adoption in high-performance propulsion systems. One of the major issues is ignition delay, which occurs due to the relatively slow initiation of combustion between the oxidizer and the solid fuel surface.

Another challenge is the low fuel regression rate, which restricts the rate at which fuel is consumed and consequently limits the thrust output of the engine. Additionally, hybrid rockets may experience combustion instabilities and incomplete mixing between fuel vapour and oxidizer, resulting in reduced combustion efficiency.

These challenges motivate the exploration of advanced techniques to enhance combustion characteristics and improve ignition reliability in hybrid rocket engines.

1.6 Plasma-Assisted Combustion Concept

Plasma-assisted combustion has emerged as a promising technique to enhance ignition and combustion processes in propulsion systems. Plasma is an ionized gas consisting of electrons, ions, and neutral species that interact strongly with electric and magnetic fields. When introduced into a combustible mixture, energetic electrons collide with fuel and oxidizer molecules, producing highly reactive radicals such as O, H, and OH, along with excited species, which accelerate chemical reactions, reduce ignition delay, and improve flame stabilization.

As shown in Fig. 1.2. plasma discharge influences combustion through multiple coupled mechanisms. It leads to temperature increase (thermal effect), generation of radicals and excited species (kinetic effect), and formation of fuel fragments and ionic wind (transport effect) that enhance mixing. These processes collectively improve reaction rates, fuel decomposition, and mass transfer within the combustion zone. The combined action of thermal, kinetic, and transport effects results in significant combustion enhancement in terms of efficiency, stability, and ignition reliability.

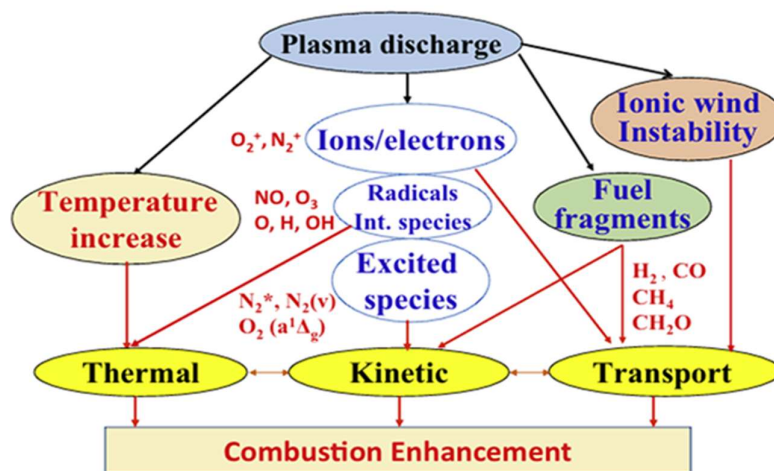


Fig. 1.2. Four Combustion enhancement pathways of plasma assisted combustion

Among the various plasma generation techniques, nanosecond pulsed plasma discharges have emerged as an effective method for enhancing ignition and combustion processes in

propulsion systems. In this approach, extremely short-duration high-voltage pulses (typically in the nanosecond range) are applied across electrodes to generate a non-equilibrium plasma discharge. Under these conditions, electrons are rapidly accelerated to high energies due to the strong electric field, while the bulk gas temperature remains comparatively low. This non-equilibrium nature allows efficient electron-impact dissociation and excitation of molecules such as O_2 and N_2 , leading to the formation of highly reactive radicals (O, H, OH) and electronically excited species that significantly accelerate combustion reactions.

A critical parameter governing the characteristics of such plasma discharges is the reduced electric field strength (E/N), where E represents the electric field intensity and N denotes the number density of neutral particles. High values of E/N enable electrons to gain sufficient kinetic energy between collisions, promoting ionization, dissociation, and excitation processes. As a result, a large concentration of reactive species is generated within a very short time scale, which enhances chemical reaction rates and reduces ignition delay in combustible mixtures.

As shown in Fig. 1.3. experimental studies typically employ a controlled discharge chamber equipped with high-voltage electrodes connected to a pulsed power generator. Additional components such as back-current and direct-current shunts are used for electrical measurements, while diagnostic tools including high-speed cameras, photomultipliers, and monochromators are utilized to capture plasma evolution and analyze emission spectra. The generated plasma streamer discharge propagates through the gas mixture, initiating chemical reactions primarily through electron-impact mechanisms and radical formation.

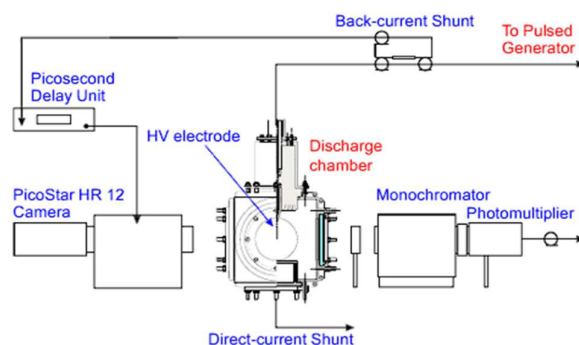


Fig. 1.3. Experimental setup for investigating nanosecond streamer discharge plasma generation.

The interaction between the plasma and the fuel–oxidizer mixture leads to rapid pre-conditioning of the reactants, thereby improving ignition characteristics and combustion stability. In hybrid rocket propulsion systems, the application of nanosecond pulsed plasma discharges can significantly enhance ignition reliability, flame holding, and combustion efficiency, particularly under conditions where conventional ignition methods may be less effective.

However, it is important to note that in the present study, the detailed plasma discharge physics, including electromagnetic effects and plasma kinetics, is not explicitly modelled within the CFD framework. Instead, the influence of plasma-assisted ignition is incorporated indirectly through modified combustion conditions derived from thermochemical (RPA) analysis, such as increased effective reaction rates or improved ignition characteristics. This approach allows the investigation of plasma-assisted effects on the internal flow and performance of the hybrid rocket nozzle using classical CFD methods, without introducing the complexity of fully coupled plasma–flow simulations.

In practical implementations, nanosecond pulsed plasma discharges are generated within the combustion chamber region, where the plasma interacts directly with the propellant mixture to initiate and sustain combustion. Thus, plasma in this study acts as an ignition and combustion enhancement mechanism, while the thrust generation continues to be governed by conventional chemical rocket gas dynamics.

1.7 Problem Statement

Although hybrid rockets offer advantages such as safety, simplicity, and controllability, their performance is often limited by slow ignition processes, low regression rates, and incomplete combustion. Plasma-assisted combustion has been proposed as a potential solution to these challenges, as it can generate reactive species that accelerate chemical reactions and improve ignition reliability.

However, the integration of plasma-assisted combustion concepts with hybrid rocket thrust chamber design and flow expansion behaviour has not been extensively investigated through computational fluid dynamics (CFD) analysis. Understanding the interaction between combustion conditions and nozzle flow characteristics is essential for evaluating the feasibility of plasma-assisted hybrid propulsion systems.

1.8 Aim of the Study

The aim of this study is to design and numerically investigate a hybrid rocket thrust chamber and converging–diverging nozzle system and to evaluate the influence of plasma-assisted combustion conditions using computational fluid dynamics simulations.

1.9 Objectives of the Study

The main objectives of this project are:

- To design the geometry of a hybrid rocket thrust chamber and converging–diverging nozzle.
- To perform thermochemical analysis of the combustion process to determine combustion properties.
- To develop a CFD model of the hybrid rocket nozzle using ANSYS Fluent.
- To analyze compressible flow characteristics such as pressure, temperature, velocity, and Mach number within the nozzle.
- To investigate the influence of plasma-assisted combustion conditions on the flow behaviour and propulsion characteristics.

1.10 Scope and Limitations

This study focuses on the numerical design and analysis of a hybrid rocket thrust chamber and nozzle system using computational fluid dynamics techniques. The investigation includes geometry development, mesh generation, and compressible flow simulation to evaluate the behaviour of combustion gases inside the rocket nozzle.

However, certain limitations are present in the current study. The plasma-assisted combustion process is represented using simplified modelling assumptions rather than detailed plasma chemistry simulations. The analysis is performed using a two-dimensional axisymmetric CFD model to reduce computational cost, and experimental validation of the design is beyond the scope of the present work.

CHAPTER II

LITERATURE REVIEW

Plasma-assisted combustion has emerged as a promising technique for improving ignition characteristics and combustion efficiency in propulsion systems. Hybrid rocket engines combine the simplicity of solid rockets with the controllability of liquid propulsion systems; however, they suffer from several limitations including ignition delay, low fuel regression rates, and incomplete mixing between oxidizer and fuel vapours. Recent studies have investigated the application of plasma-assisted ignition techniques to enhance combustion stability and accelerate reaction kinetics in propulsion systems.

Plasma discharges generate highly energetic electrons and reactive radicals that significantly accelerate chemical reaction pathways. These properties make plasma-assisted combustion an attractive approach for improving ignition reliability in hybrid rocket engines. In many plasma-assisted ignition systems, the discharge is produced between electrode pairs connected to high-voltage pulsed power supplies, creating localized plasma regions within the combustible mixture.

2.1. Studies on Plasma–Interaction in Hybrid Rocket Combustion

Boyd et al. [1] developed numerical methods for simulating electromagnetic propulsion systems and demonstrated the capability of CFD techniques to capture plasma–flow interactions and ion acceleration processes. Although the study focused on electric propulsion systems, the numerical methods provide valuable insights into the modelling of plasma interactions within propulsion systems.

Kim et al. [2] studied plasma-assisted flame stabilization in combustion systems and demonstrated that nanosecond pulsed plasma discharges can significantly improve flame stability and reduce ignition delay times. These findings indicate that plasma-assisted techniques can improve combustion stability in propulsion systems operating under challenging conditions.

In addition to experimental studies, numerical Modelling approaches have also been employed to understand plasma–flow interactions and combustion behaviour in propulsion systems.

Starikovskiy and Aleksandrov et al. [3] reviewed the mechanisms of plasma-assisted ignition and reported that nanosecond pulsed discharges are highly effective in generating radicals and initiating combustion reactions. Their work demonstrated that plasma-assisted ignition can significantly reduce ignition delay and improve flame stability under various operating conditions. These plasma discharges are typically generated using electrode-based pulsed power systems where high-voltage nanosecond pulses create streamer discharges within the combustion mixture.

Ju and Sun et al. [4] investigated plasma-assisted combustion mechanisms and demonstrated that non-equilibrium plasma can significantly improve ignition performance and extend flammability limits in hydrocarbon fuels. Their research showed that plasma-generated radicals play a crucial role in accelerating combustion reactions, particularly during the initial ignition phase.

Nakagawa et al. [5] investigated the use of a plasma torch igniter for hybrid rocket engines. The study demonstrated that plasma torch ignition systems can successfully ignite hybrid rocket fuels under both atmospheric and high-pressure conditions. The plasma torch generated localized high temperatures that enabled rapid ignition without the need for traditional ignition systems. The plasma torch generated localized high temperatures that enabled rapid ignition without the need for traditional ignition systems. Such plasma ignition devices typically employ electrode configurations to sustain high-energy discharge within the combustion region.

Karabeyoglu et al. [6] investigated the combustion behaviour and regression characteristics of hybrid rocket fuels. Their research demonstrated that paraffin-based fuels can significantly improve fuel regression rates due to the formation of liquid droplets that enhance turbulent mixing within the combustion boundary layer. The study provided important insights into hybrid rocket combustion physics and highlighted the importance of improving mixing and ignition processes to enhance propulsion performance.

Chavda and Kulkarni et al. [7] provided a comprehensive review of plasma-assisted combustion technologies in propulsion systems. The study highlighted that non-equilibrium plasmas are capable of generating reactive radicals that enhance ignition, stabilize flames, and improve combustion efficiency in various aerospace applications.

Andersson et al. [8] conducted numerical simulations of combustion processes in hybrid rocket engines using a GOX–HDPE propellant combination. The results showed that combustion efficiency is strongly influenced by propellant mixing and heat transfer mechanisms within the combustion chamber. The study emphasized the importance of advanced ignition techniques and improved combustion Modelling for optimizing hybrid rocket performance.

Zhang et al. [9] performed numerical simulations of plasma-enhanced combustion in Rocket-Based Combined Cycle (RBCC) engines using URANS Modelling techniques. The study showed that plasma-assisted combustion improved fuel–air mixing and enhanced combustion stability, demonstrating the effectiveness of plasma-assisted ignition in high-speed propulsion systems.

2.2 Conclusions from Literature Review

The reviewed literature indicates that hybrid rocket propulsion systems provide several advantages, including operational safety, structural simplicity, and controllability when compared to conventional solid and liquid rocket engines. However, hybrid rockets still face significant challenges related to ignition delay, low fuel regression rates, and incomplete mixing between fuel vapours and oxidizer streams, which can limit combustion efficiency and propulsion performance [1], [2]. Previous studies have shown that improving mixing mechanisms and combustion processes is essential for enhancing hybrid rocket performance. Plasma-assisted combustion has emerged as a promising approach for addressing these limitations, as non-equilibrium plasma discharges generate highly energetic electrons that produce reactive radicals such as O, H, and OH through electron-impact dissociation processes [3], [4], [5]. These radicals significantly accelerate chemical reaction pathways, reduce ignition delay times, and improve flame stabilization under various operating conditions. Experimental studies on plasma ignition systems have demonstrated the feasibility of plasma-based ignition in hybrid rocket engines, showing that plasma torch igniters can produce localized high-temperature regions capable of initiating reliable combustion without conventional pyrotechnic systems [6]. Furthermore, research on nanosecond pulsed plasma discharges indicates that plasma activation can substantially enhance flame stability and combustion efficiency in high-speed propulsion environments

[7]. In addition to experimental investigations, numerical studies have demonstrated that computational fluid dynamics is an effective tool for analysing complex flow behaviour, compressible flow expansion, and combustion characteristics within rocket propulsion systems [8], [9]. These findings collectively suggest that integrating plasma-assisted ignition concepts with CFD-based analysis of hybrid rocket thrust chambers can provide deeper insights into combustion enhancement mechanisms and propulsion performance. Therefore, the present study focuses on the design and CFD validation of a hybrid rocket thrust chamber and investigates the potential influence of plasma-assisted combustion conditions on the internal flow characteristics of the propulsion system.

2.3 LITERATURE GAPS

- Most studies on hybrid rocket propulsion focus primarily on fuel regression behaviour and combustion efficiency, while limited attention has been given to improving ignition reliability using advanced techniques such as plasma-assisted ignition.
- A large portion of plasma-assisted combustion research has been conducted for gas turbines, scramjets, and internal combustion engines, with comparatively fewer studies addressing its application in hybrid rocket propulsion systems.
- Previous investigations on plasma ignition systems mainly emphasize experimental ignition methods, whereas the interaction between plasma-assisted combustion conditions and the internal flow characteristics of hybrid rocket nozzles has not been extensively studied.
- Limited research has been conducted on analysing the compressible flow behaviour in hybrid rocket thrust chambers under plasma-assisted combustion conditions.
- Experimental investigation of plasma-assisted hybrid propulsion systems is often complex and expensive; therefore, numerical simulation approaches such as CFD are required for preliminary analysis and performance prediction.
- There is a lack of studies that integrate hybrid rocket thrust chamber design, thermochemical analysis, and CFD-based flow analysis within a single framework to evaluate propulsion performance.
- The influence of plasma-assisted combustion on key flow parameters such as pressure distribution, temperature variation, velocity fields, and Mach number development within rocket nozzles has not been thoroughly investigated.

CHAPTER III

METHODOLOGY

3.1 Introduction

This chapter describes the methodology adopted for the design and computational analysis of the hybrid rocket thrust chamber and nozzle system. The objective of this chapter is to explain the procedures followed in developing the propulsion system geometry, performing thermochemical analysis, and conducting computational fluid dynamics (CFD) simulations to analyze the flow characteristics inside the rocket nozzle.

The methodology includes several key stages such as conceptual design of the hybrid rocket propulsion system, development of the thrust chamber and nozzle geometry, creation of the computational domain, mesh generation, and numerical simulation using ANSYS Fluent. The compressible flow behaviour inside the rocket nozzle is analysed by solving the governing equations of mass, momentum, and energy conservation. In addition, the potential influence of plasma-assisted combustion conditions on the internal flow characteristics of the propulsion system is investigated through numerical modeling.

The workflow followed in this study integrates theoretical rocket propulsion principles with numerical simulation techniques to evaluate the performance of the hybrid rocket propulsion system.

3.2. Design Approach

The present study focuses on the design and computational analysis of a hybrid rocket thrust chamber and converging–diverging nozzle to investigate the internal flow characteristics under plasma-assisted combustion conditions. The methodology adopted in this study consists of several sequential steps including theoretical analysis, geometry development, computational fluid dynamics (CFD) simulation, and flow analysis.

Initially, the propulsion system configuration and nozzle geometry were designed based on classical rocket propulsion theory. The combustion properties and thermodynamic parameters were estimated using thermochemical analysis. The geometric model of the hybrid rocket thrust chamber was then developed using CAD software. Subsequently, the computational domain was created and discretized using an appropriate mesh. Finally, numerical simulations were performed using ANSYS Fluent to analyze the compressible flow characteristics inside the rocket nozzle.

3.3. Hybrid Rocket Engine Configuration

A hybrid rocket engine consists of a solid fuel grain and a liquid or gaseous oxidizer. The oxidizer is injected into the combustion chamber where it reacts with the vaporized fuel surface to produce high-temperature combustion gases. These gases expand through a converging–diverging nozzle to generate thrust.

The main components of the hybrid rocket propulsion system include:

- Oxidizer injection system
- Combustion chamber containing the solid fuel grain
- Converging–diverging nozzle
- Exhaust section

The combustion chamber provides the region where the fuel and oxidizer react to produce high-pressure combustion gases. The nozzle accelerates these gases from subsonic velocity in the chamber to supersonic velocity at the exit, converting thermal energy into kinetic energy.

3.4. Rocket Nozzle Design Theory

The performance of a rocket engine is strongly influenced by the design of the converging–diverging nozzle. The nozzle accelerates combustion gases and converts thermal energy into kinetic energy to produce thrust.

3.4.1 Rocket Thrust Equation

The thrust generated by a rocket engine is given by:

$$F = \dot{m}V_e + (P_e - P_a)A_e$$

where:

F = thrust (N)

$\dot{m}$ = mass flow rate (kg/s)

V_e = exhaust velocity (m/s)

P_e = exit pressure (Pa)

P_a = ambient pressure (Pa)

A_e = exit area (m²)

3.4.2 Area–Mach Number Relation

The area–Mach number relationship is given by:

$$\frac{A}{A^*} = \frac{1}{M} \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M^2 \right) \right]^{\frac{\gamma + 1}{2(\gamma - 1)}}$$

where:

A = local cross-sectional area

A^* = throat area

M = Mach number

γ = specific heat ratio

3.4.3 Choked Flow Condition

For a rocket nozzle to operate efficiently, the flow must reach sonic velocity at the throat. This condition is known as choked flow and occurs when the Mach number at the throat equals one.

$$M = 1 \quad \text{at the throat}$$

At this condition, the mass flow rate through the nozzle becomes independent of downstream pressure.

3.5 Thermochemical Analysis Using RPA

Thermochemical analysis plays an essential role in rocket engine design because it provides the combustion properties required for performance evaluation and numerical simulations. In rocket propulsion systems, the combustion of fuel and oxidizer produces high-temperature gases whose thermodynamic properties determine the expansion behaviour within the nozzle and ultimately influence thrust generation.

In the present study, thermochemical analysis of the hybrid rocket propellant combination was performed using Rocket Propulsion Analysis (RPA) software. RPA is widely used in

rocket propulsion research to calculate equilibrium combustion properties based on propellant composition, chamber pressure, and mixture ratio.

The software computes important parameters such as:

Table 3.1. Thermochemical Properties from RPA Analysis

Parameter	Value
Chamber Pressure	15 bar
Combustion Temperature	~4211K
Specific Heat Ratio (γ)	1.255
Molecular Weight	225.89 kg/kmol
Gas Constant	0.321 kJ/kg·K
Specific Heat (C_p)	6.38 kJ/kg·K
Exit Mach Number	2.67

These thermodynamic properties are essential inputs for nozzle design and CFD simulations because they determine the compressible flow behaviour of the combustion gases inside the rocket engine.

The combustion properties obtained from the RPA analysis were used to define the operating conditions for the CFD simulation of the hybrid rocket nozzle. In particular, the combustion chamber temperature and gas properties were used as reference values for the inlet conditions of the numerical model.

3.6 Geometry Design

The geometry of the hybrid rocket thrust chamber and converging–diverging nozzle was developed based on analytical design calculations derived from classical rocket propulsion theory. The objective of the design was to ensure efficient expansion of high-temperature combustion gases to generate supersonic exhaust flow while maintaining geometric simplicity suitable for CFD analysis.

As shown in Fig. 3.1. the complete three-dimensional model of the thrust chamber and nozzle was created using SolidWorks. The geometry consists of a cylindrical combustion chamber followed by a converging section, a throat, and a diverging nozzle. For computational efficiency, the 3D geometry was subsequently converted into a two-dimensional axisymmetric model, which accurately represents the flow physics due to the rotational symmetry of the configuration.

The key geometric parameters used in the design are summarized in Table 3.2. These parameters were selected to achieve the required expansion characteristics and flow acceleration through the nozzle.

The selected expansion ratio (5) ensures adequate conversion of thermal energy into kinetic energy, enabling supersonic flow at the nozzle exit. The throat diameter governs the mass flow rate under choked flow conditions, while the divergent angle (11°) is chosen to minimize flow separation and ensure efficient expansion. The convergent angle (30°) facilitates smooth acceleration of subsonic flow toward the throat.

Table 3.2. Design Parameters of the Hybrid Rocket

Parameter	Value
Chamber Diameter	16 mm
Chamber Length	150 mm
Throat Diameter	8 mm
Exit Diameter	17.9 mm
Expansion Ratio	5
Nozzle Length	25.6 mm
Convergent Angle	30°
Divergent Angle	11°

The designed geometry represents a simplified hybrid rocket thrust chamber suitable for computational analysis of compressible flow behaviour.

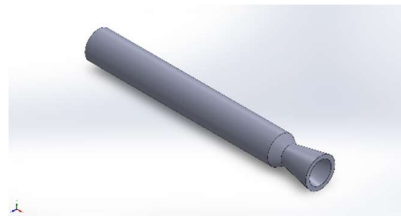


Fig. 3.1. SolidWorks 3D geometry

3.7 Computational Domain

As shown in Fig. 3.2, the rocket nozzle was modelled using a two-dimensional axisymmetric domain to reduce computational complexity while maintaining physical accuracy. Since the geometry is rotationally symmetric about the central axis, this approach accurately represents the three-dimensional flow behavior with significantly lower computational cost.



Fig. 3.2. Axisymmetric computational domain of rocket nozzle.

The computational domain includes the combustion chamber, converging section, throat, and diverging section, ensuring complete representation of the flow expansion process within the nozzle. As illustrated in Fig. 3.3, appropriate named selections were defined for boundary condition implementation, including pressure inlet at the chamber entry, pressure outlet at the nozzle exit, Farfield, nozzle wall, and the axis of symmetry. These boundary definitions ensure accurate application of flow conditions in the CFD simulation.

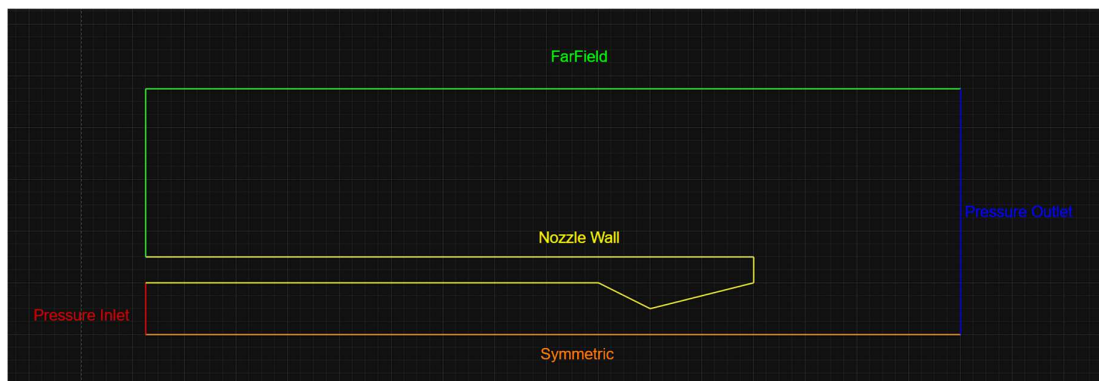


Fig. 3.3. Named boundary regions in the axisymmetric rocket nozzle model

3.8 Mesh Generation

The computational domain was discretized using a structured quadrilateral-dominant mesh, as shown in Fig. 3.4. (Mesh structure of 2D axisymmetric hybrid rocket), to ensure accurate resolution of compressible flow within the nozzle.

A face meshing approach was applied by dividing all four faces of the domain systematically. Additionally, edge sizing was implemented along five key edges, with the number of divisions varying from 50 to 200 depending on the region. Finer mesh (≈ 150 –200 divisions) was used near the throat region to capture strong pressure and velocity gradients, while comparatively coarser mesh was used in regions with relatively uniform flow.

Mesh quality was verified to maintain acceptable skewness and aspect ratio, ensuring numerical stability and solution accuracy.

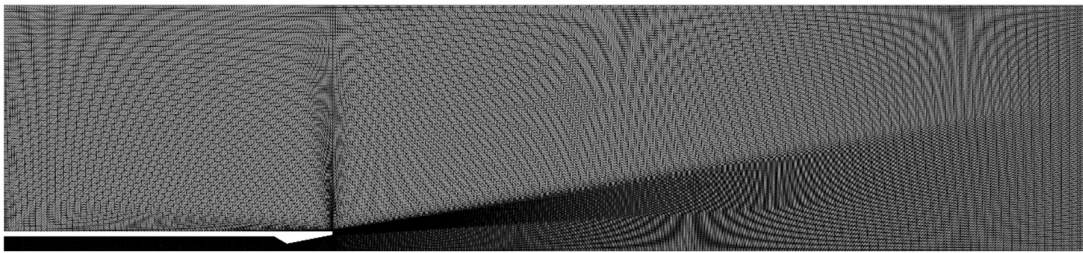


Fig. 3.4. Mesh structure of 2D Axisymmetric hybrid rocket

3.9 Mesh Independence Study

A mesh independence study was performed to ensure that the numerical results were not affected by the grid resolution. Simulations were conducted using different mesh densities, and the variation in key flow parameters such as Mach number and pressure distribution was analysed.

The results indicated that beyond a certain mesh density, further refinement produced negligible changes in the solution. Therefore, an optimal mesh size was selected to balance computational accuracy and simulation efficiency.

3.10 Boundary Conditions

The CFD simulations were performed using appropriate boundary conditions to accurately represent the operating conditions of the hybrid rocket engine. A pressure inlet boundary condition was applied at the combustion chamber inlet, while a pressure outlet was specified at the nozzle exit. The chamber and nozzle walls were modelled using a no-slip wall condition, and an axis boundary condition was applied along the centreline to represent the axisymmetric configuration.

The specific boundary condition parameters used in the simulation are summarized in Table 3.3 (Boundary Condition Parameters). A total pressure of 15 bar and a temperature of 4211 K, obtained from thermochemical analysis, were applied at the inlet to represent the combustion conditions. At the nozzle exit, an ambient pressure of 1 atm was specified to simulate expansion into atmospheric conditions.

A density-based solver with a compressible flow formulation was employed to accurately capture high-speed flow behaviour, including compressibility effects and potential supersonic expansion within the nozzle. These boundary conditions ensure a realistic representation of the internal flow field and are suitable for analyzing the performance characteristics of the hybrid rocket nozzle.

The following boundary conditions were applied:

- Pressure inlet at the combustion chamber inlet
- Pressure outlet at the nozzle exit
- No-slip wall condition for the chamber and nozzle walls
- Axis boundary condition along the centreline

Table 3.3. Boundary Conditions Parameters

Parameter	Value
Total Pressure	15 bar
Temperature	4211 K
Outlet Pressure	1 atm

3.11 Numerical Simulation Setup

The numerical simulations were performed using ANSYS Fluent. A density-based solver was used to model compressible flow conditions within the rocket nozzle. The governing equations for mass, momentum, and energy conservation were solved using finite volume methods.

The energy equation was enabled to account for temperature variations and thermal effects within the flow field. Appropriate discretization schemes were selected to ensure numerical stability and accurate solution convergence.

3.12 Plasma-Assisted Combustion Representation

In the present study, plasma-assisted combustion effects were represented using a simplified modelling approach. Instead of directly simulating plasma chemistry, the influence of plasma-assisted ignition was approximated by modifying the combustion temperature conditions within the chamber.

This approach allows the investigation of how enhanced combustion conditions influence the compressible flow behaviour within the rocket nozzle while maintaining computational efficiency.

3.13 Plasma Ignition Electrode Configuration

Plasma-assisted ignition is considered in this study as a potential method for enhancing ignition reliability in hybrid rocket propulsion systems. Plasma ignition systems generate high-energy electrons through electrical discharge between electrodes, which produce reactive radicals capable of accelerating combustion reactions.

In the present conceptual configuration, a nanosecond pulsed high-voltage discharge is generated between a pair of electrodes positioned inside the combustion chamber near the propellant mixing region. When a nanosecond voltage pulse is applied across the electrodes, a localized plasma discharge is formed within the oxidizer–fuel mixture. The energetic electrons generated during the discharge interact with molecular species and produce reactive radicals such as O, H, and OH, which significantly accelerate the chemical reaction pathways responsible for ignition.

A pin–pin electrode arrangement is considered in the present design due to its ability to generate stable and localized plasma discharges within a confined combustion environment. The electrodes are placed upstream of the converging section to ensure effective interaction between the plasma discharge and the propellant mixture without disturbing the primary expansion of gases through the nozzle.

It should be noted that the present study does not directly simulate plasma discharge physics. Instead, the influence of plasma-assisted ignition is represented through modified thermochemical conditions derived from Rocket Propulsion Analysis (RPA). This simplified approach allows the investigation of the potential impact of plasma-assisted ignition on the internal flow characteristics of the hybrid rocket nozzle using computational fluid dynamics simulations.

3.14 Electrode Material Selection

Electrode material selection is a critical aspect of plasma-assisted ignition systems, as electrodes are subjected to high temperatures, intense electric fields, and plasma-induced erosion during discharge events. Therefore, the selected material must exhibit a high melting temperature, good electrical conductivity, and strong resistance to thermal and plasma degradation to ensure durability and stable operation.

Common electrode materials used in plasma ignition systems include tungsten, copper, and copper–tungsten composite alloys. As summarized in Table 3.4 (Comparison of Electrode Materials for Plasma Discharge Applications), tungsten has an extremely high melting point of 3422°C and excellent resistance to plasma erosion, making it highly suitable for high-temperature environments, although its electrical conductivity is moderate. Copper, on the other hand, offers very high electrical conductivity, which is beneficial for efficient current flow, but its relatively low melting point (1085°C) limits its performance under intense thermal conditions. Copper–tungsten composites provide a balanced combination of high thermal resistance ($\approx 3000^\circ\text{C}$), good electrical conductivity, and high erosion resistance, making them a preferred choice in applications requiring both durability and efficient electrical performance.

Table 3.4. Comparison of Electrode Materials for Plasma Discharge Applications

Material	Melting Point(°C)	Electrical Conductivity	Plasma Erosion Resistance
Tungsten	3422	Moderate	Very High
Copper	1085	Very High	Low
Copper-Tungsten	~ 3000	High	High

CHAPTER IV

RESULTS AND DISCUSSION

This chapter presents the computational results obtained from the CFD simulation of the hybrid rocket thrust chamber and converging–diverging nozzle. The numerical analysis was performed to investigate the compressible flow behaviour of the combustion gases inside the propulsion system. The simulation results provide detailed insights into the distribution of velocity, pressure, temperature, and Mach number within the rocket nozzle.

The results obtained from the simulation are analysed to evaluate the performance of the designed nozzle and to understand the flow acceleration from subsonic to supersonic conditions. The influence of the nozzle geometry on the expansion of combustion gases and the development of supersonic flow at the nozzle exit is also discussed.

4.1 Flow Field Distribution in the Hybrid Rocket Nozzle

(Without plasma Interaction)

4.1.1 Velocity Distribution

Figure. 4.1. presents the velocity contour obtained from the CFD simulation of the hybrid rocket thrust chamber and converging–diverging nozzle under baseline conditions without plasma interaction. The contour illustrates the acceleration of combustion gases from the combustion chamber toward the nozzle exit.

Inside the combustion chamber, the flow velocity remains relatively low due to the large cross-sectional area and high static pressure. As the gases enter the converging section, the velocity gradually increases due to the reduction in flow area.

At the nozzle throat, the flow reaches sonic conditions ($\text{Mach} \approx 1$), indicating choked flow. Beyond the throat, the gases expand through the diverging section and accelerate rapidly, reaching velocities of approximately 2700–2750 m/s near the nozzle exit.

This acceleration results from the conversion of thermal energy of the high-temperature combustion gases into kinetic energy during expansion. The velocity distribution confirms that the designed converging–diverging nozzle effectively accelerates the flow to high exhaust velocities required for efficient rocket thrust generation.

The observed velocity profile is consistent with classical compressible nozzle flow theory, where the expansion of high-temperature gases through a converging-diverging nozzle produces supersonic exhaust required for rocket propulsion.

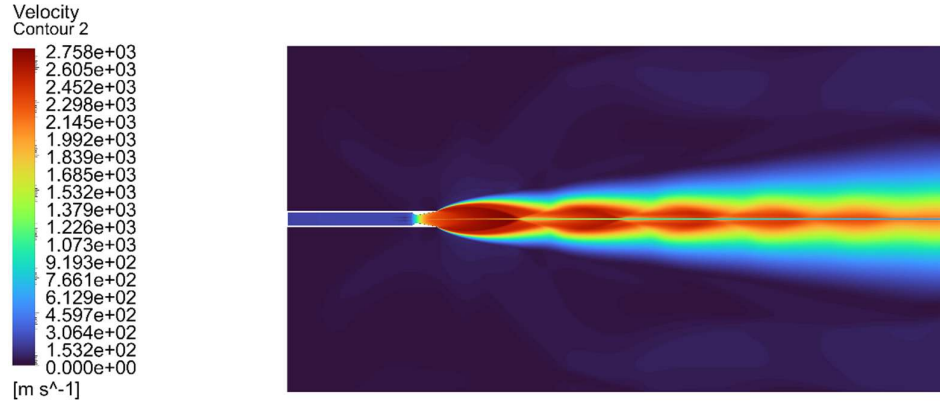


Fig. 4.1. Velocity contour of the hybrid rocket nozzle.

4.1.2 Pressure Distribution

Figure. 4.2. shows the static pressure contour obtained from the CFD simulation of the hybrid rocket thrust chamber and converging-diverging nozzle under baseline operating conditions. The pressure distribution illustrates the expansion of combustion gases from the high-pressure combustion chamber to the low-pressure exhaust region.

Inside the combustion chamber, the pressure remains relatively high due to the confinement of the combustion gases. As the flow enters the converging section of the nozzle, the pressure gradually decreases while the velocity increases. At the nozzle throat, the pressure drops significantly as the flow reaches sonic conditions ($\text{Mach} \approx 1$) and the mass flow becomes choked.

Beyond the throat, the pressure decreases rapidly within the diverging section due to the expansion of the gases and the acceleration to supersonic velocities. Near the nozzle exit, the pressure approaches the ambient level as the exhaust jet expands into the surrounding domain.

The pressure contour clearly demonstrates the conversion of pressure energy into kinetic energy during the expansion process. The observed pressure variation is consistent with isentropic compressible flow behaviour in converging-diverging rocket nozzles, confirming the proper operation of the designed propulsion system.



Fig. 4.2. Pressure contour of the rocket nozzle.

4.1.3 Temperature Distribution

Figure. 4.3. presents the temperature contour obtained from the CFD simulation of the hybrid rocket thrust chamber and converging–diverging nozzle under baseline operating conditions. The temperature distribution illustrates the thermal behaviour of the combustion gases as they expand through the nozzle.

The highest temperature occurs within the combustion chamber region, where the combustion of the fuel–oxidizer mixture produces high thermal energy. In the present simulation, the chamber temperature reaches approximately 4200 K, which is consistent with the thermochemical predictions obtained from the RPA analysis.

As the gases enter the converging–diverging nozzle, the temperature gradually decreases due to the expansion of the flow. A significant temperature reduction occurs downstream of the throat as the gases accelerate to supersonic velocities in the diverging section. This temperature drop occurs because the internal thermal energy of the gas is converted into kinetic energy during the expansion process.

The temperature contour clearly illustrates the cooling of the exhaust gases along the nozzle length, which is a characteristic behaviour of adiabatic compressible flow expansion in rocket propulsion systems.

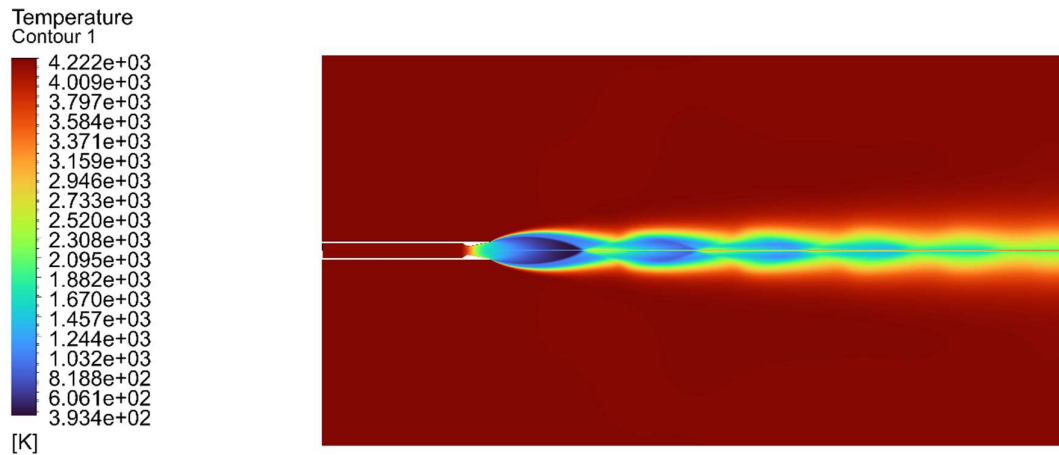


Fig. 4.3. Temperature contour of the rocket nozzle

4.1.4 Mach Number Distribution

Figure. 4.4. shows the Mach number contour obtained from the CFD simulation of the hybrid rocket thrust chamber and converging–diverging nozzle under baseline operating conditions. The contour clearly illustrates the transition of the flow from subsonic conditions in the combustion chamber to supersonic conditions at the nozzle exit.

Inside the combustion chamber, the flow remains subsonic ($\text{Mach} < 1$) due to the relatively large cross-sectional area and high pressure of the combustion gases. As the flow approaches the converging section of the nozzle, the Mach number gradually increases as the velocity rises.

At the nozzle throat, the Mach number reaches unity ($\text{Mach} \approx 1$), indicating the occurrence of choked flow, which fixes the mass flow rate through the nozzle. Downstream of the throat, the gases expand rapidly in the diverging section and accelerate to supersonic velocities.

The Mach number contour shows that the exhaust flow reaches values of approximately Mach 2.5–2.7 near the nozzle exit, which is consistent with the theoretical predictions obtained from the thermochemical analysis using RPA.

The Mach number distribution confirms the correct operation of the converging–diverging nozzle, where the expansion of high-temperature combustion gases produces supersonic exhaust required for efficient rocket propulsion.

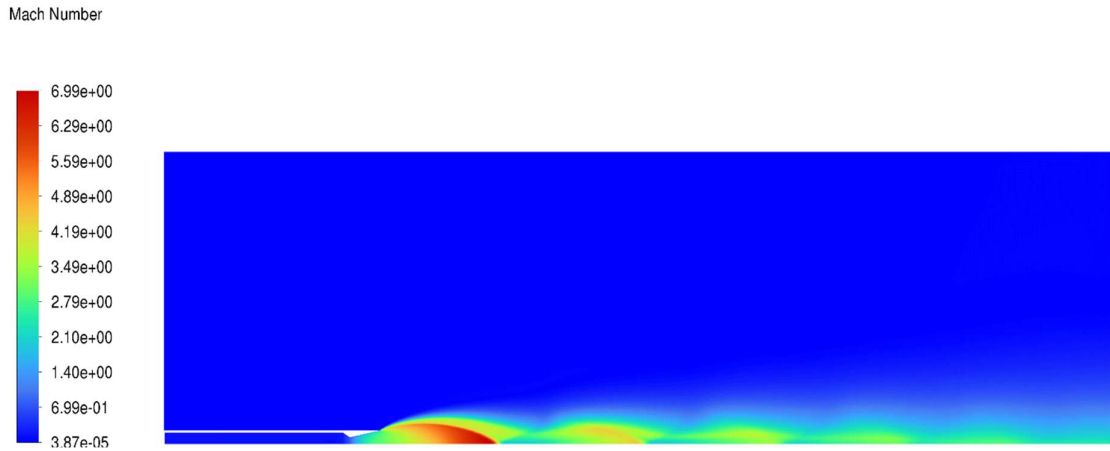


Fig. 4.4. Mach number contour of the hybrid rocket nozzle.

4.2 Streamline Visualization

Figure. 4.5. presents the velocity streamlines obtained from the CFD simulation of the hybrid rocket nozzle under baseline operating conditions. The streamlines illustrate the flow path of the combustion gases as they move from the combustion chamber through the converging–diverging nozzle.

The streamlines show that the flow accelerates smoothly from the combustion chamber toward the nozzle exit. As the gases pass through the converging section, the streamlines become closer, indicating an increase in flow velocity due to the reduction in cross-sectional area. Beyond the throat, the streamlines expand along the diverging section as the gases accelerate to supersonic velocities.

The streamline pattern indicates stable and attached flow throughout the nozzle without significant flow separation or recirculation regions. This behaviour confirms that the designed nozzle geometry provides efficient expansion of the combustion gases and supports stable supersonic exhaust flow.

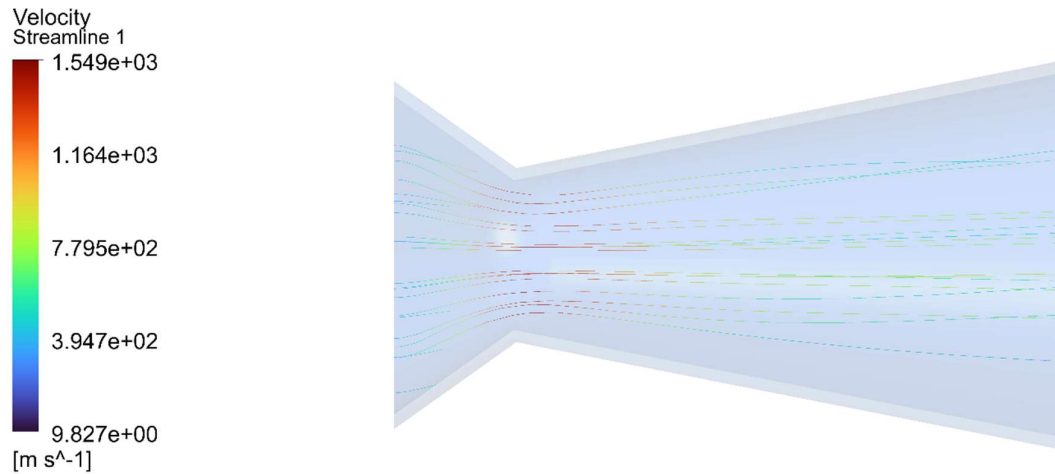


Fig. 4.5. Streamlines of flow inside the rocket nozzle.

4.4 Plasma-Assisted Combustion Comparison

To investigate the potential influence of plasma-assisted combustion, an additional simulation case was considered where the combustion chamber temperature was increased to represent plasma-enhanced chemical reactions.

The increase in combustion temperature used in the plasma-assisted case represents the additional energy deposition that would be produced by nanosecond pulsed plasma discharges within the combustion chamber.

The increased chamber temperature in the plasma-assisted case represents the additional energy deposition that would occur due to nanosecond pulsed plasma discharge within the combustion chamber.

(With Plasma Interaction)

4.4.1 Velocity Distribution

Figure. 4.6. shows the velocity contour obtained from the CFD simulation of the hybrid rocket nozzle under plasma-assisted combustion conditions. In this case, the chamber temperature was increased to represent the additional energy deposition produced by nanosecond pulsed plasma discharge within the combustion chamber.

The velocity contour shows that the flow acceleration pattern remains similar to the baseline case. Inside the combustion chamber, the flow velocity remains relatively low due to the large cross-sectional area and high pressure of the combustion gases. As the gases pass

through the converging section of the nozzle, the velocity increases due to the reduction in flow area.

At the nozzle throat, the flow reaches sonic conditions, indicating choked flow. Beyond the throat, the gases expand rapidly within the diverging section and accelerate to supersonic velocities toward the nozzle exit.

Compared to the baseline case, a slight increase in exhaust velocity can be observed in the plasma-assisted condition. The maximum velocity near the nozzle exit reaches approximately 2800–2900 m/s, which is slightly higher than the velocity obtained in the non-plasma case.

This increase occurs because the higher combustion temperature provides greater thermal energy to the flow, which is converted into kinetic energy during the expansion process.

The velocity distribution indicates that plasma-assisted combustion can enhance the expansion of combustion gases and slightly increase the exhaust velocity of the rocket nozzle.

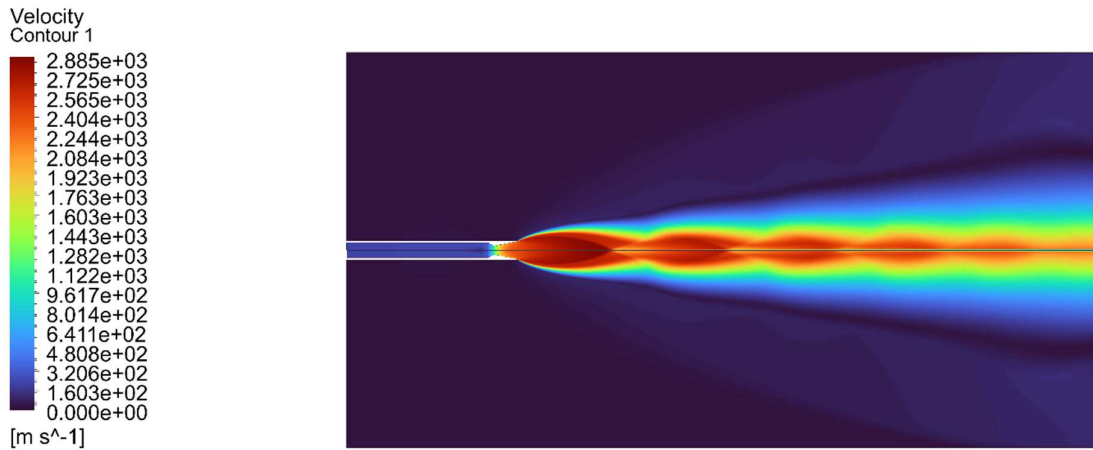


Fig. 4.6. Velocity contour with plasma

4.4.2 Pressure Distribution

Figure. 4.7. shows the static pressure contour obtained from the CFD simulation of the hybrid rocket nozzle under plasma-assisted combustion conditions. The pressure distribution illustrates the expansion of combustion gases from the high-pressure combustion chamber to the low-pressure exhaust region.

Inside the combustion chamber, the pressure remains relatively high due to the confinement of the combustion gases. As the flow enters the converging section of the nozzle, the pressure begins to decrease while the velocity increases due to the reduction in cross-sectional area.

At the nozzle throat, the pressure drops significantly as the flow reaches sonic conditions and the mass flow becomes choked. Downstream of the throat, the gases expand rapidly in the diverging section, leading to a further reduction in pressure as the flow accelerates to supersonic velocities.

The pressure distribution pattern observed in the plasma-assisted case remains similar to the baseline case. However, due to the higher chamber temperature representing plasma energy deposition, the expansion of the gases becomes slightly stronger, which contributes to increased exhaust velocity.

The pressure contour therefore confirms that the plasma-assisted case maintains the expected isentropic expansion behaviour of compressible flow within a converging–diverging rocket nozzle.

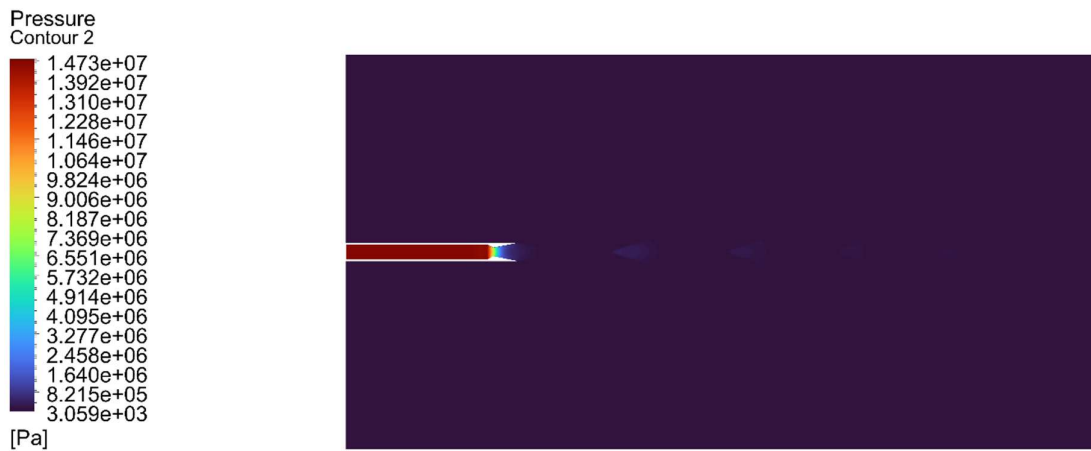


Fig .4.7. Pressure contour with plasma

4.4.3 Temperature Distribution

Figure. 4.8. shows the temperature contour obtained from the CFD simulation of the hybrid rocket nozzle under plasma-assisted combustion conditions. In this case, the combustion chamber temperature was increased to represent the additional energy deposition produced by nanosecond pulsed plasma discharge.

The temperature contour indicates that the highest temperature occurs within the combustion chamber region, where the combustion process generates significant thermal energy. In the

plasma-assisted case, the chamber temperature reaches approximately 4550–4600 K, which is higher than the baseline temperature due to the additional plasma energy input.

As the combustion gases enter the converging–diverging nozzle, the temperature gradually decreases due to the expansion of the flow. A significant temperature drop occurs downstream of the throat as the gases accelerate to supersonic velocities in the diverging section.

The reduction in temperature along the nozzle length occurs because the internal thermal energy of the gases is converted into kinetic energy during the expansion process. Compared to the baseline case, the higher chamber temperature results in slightly higher exhaust energy levels and increased expansion velocity.

The temperature distribution observed in the plasma-assisted case follows the expected adiabatic expansion behaviour of compressible flow in rocket propulsion systems.

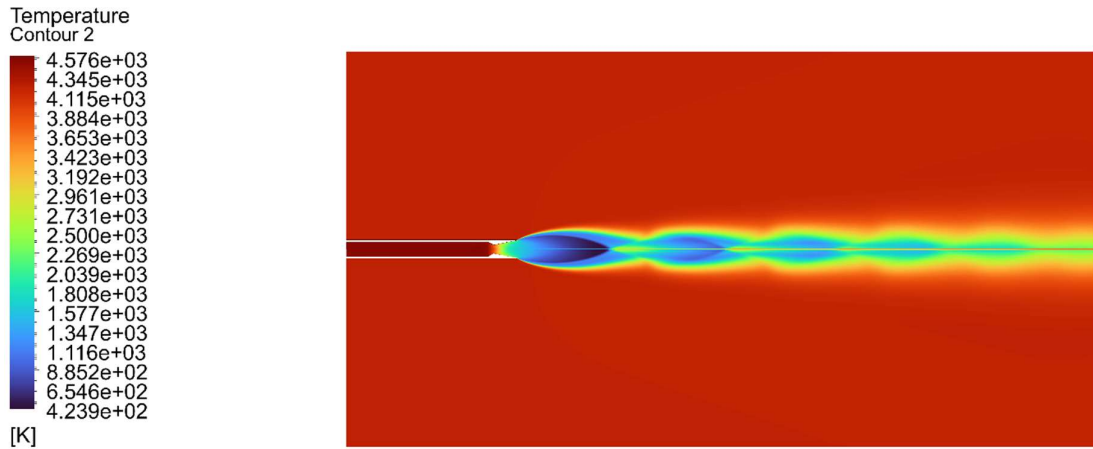


Fig. 4.8. Temperature contour with plasma

4.5.4 Mach Number Distribution

Figure. 4.9. shows the Mach number contour obtained from the CFD simulation of the hybrid rocket nozzle under plasma-assisted combustion conditions. The contour illustrates the transition of the flow from subsonic conditions in the combustion chamber to supersonic flow in the diverging section of the nozzle.

Inside the combustion chamber, the flow remains subsonic due to the relatively large cross-sectional area and high static pressure. As the gases approach the converging section of the nozzle, the Mach number gradually increases due to the rise in flow velocity.

At the nozzle throat, the flow reaches Mach 1, indicating the establishment of choked flow. Downstream of the throat, the gases expand through the diverging section and accelerate to supersonic velocities as the pressure and temperature decrease during expansion.

Compared to the baseline case, the plasma-assisted condition results in a slightly higher Mach number distribution in the divergent section of the nozzle. The additional thermal energy introduced by plasma-enhanced combustion increases the exhaust velocity, which leads to a marginal increase in the Mach number near the nozzle exit.

The Mach number distribution confirms that the nozzle maintains proper compressible flow behaviour while producing supersonic exhaust flow required for efficient rocket propulsion.

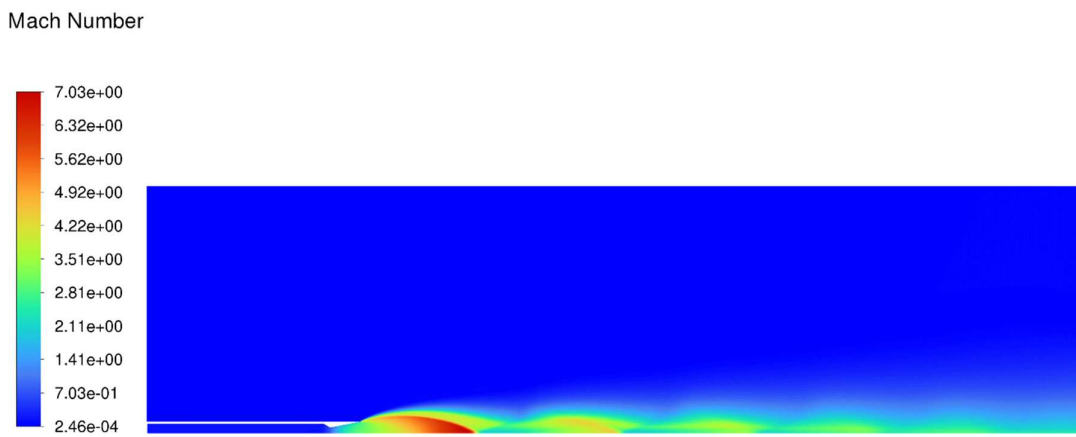


Fig. 4.9. Mach contour with plasma

4.5 Centreline Flow Property Analysis

To better understand the flow characteristics inside the rocket nozzle, the variation of key flow parameters along the centreline of the nozzle was analysed. The parameters considered include pressure, temperature, and Mach number.

The centreline plots show that the pressure decreases continuously from the combustion chamber to the nozzle exit. Similarly, the temperature decreases as the gases expand through the nozzle. The Mach number increases gradually and reaches supersonic values beyond the throat.

These trends are consistent with compressible flow theory and confirm the proper operation of the converging–diverging nozzle.

The centreline plots provide quantitative confirmation of the compressible flow behaviour observed in the contour plots, demonstrating the gradual conversion of pressure and thermal energy into kinetic energy along the nozzle length.

(Without Plasma)

4.5.1 Centreline Mach Number Variation

Figure. 4.10. presents the variation of Mach number along the nozzle centreline. The plot shows that the flow remains subsonic within the combustion chamber and increases rapidly near the throat region.

At the throat, the Mach number approaches Mach 1, confirming the establishment of choked flow. Downstream of the throat, the Mach number increases significantly as the gases expand in the diverging section, reaching supersonic values along the nozzle length.

The oscillations observed in the supersonic region correspond to shock-cell structures in the exhaust jet, which commonly occur in compressible nozzle flows due to expansion and recompression waves.

The centreline Mach number profile provides quantitative confirmation of the subsonic–sonic–supersonic transition predicted by compressible flow theory.

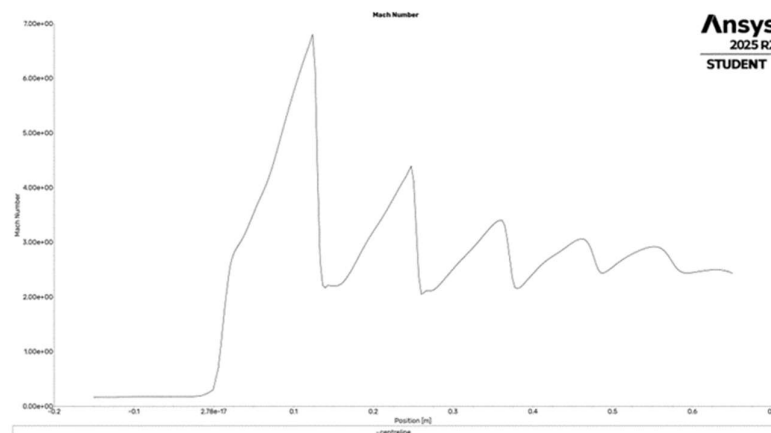


Fig. 4.10. Variation of Mach number along the nozzle axis

4.5.2 Centreline Pressure Variation

Figure. 4.11. presents the variation of static pressure along the nozzle centreline. The plot shows that the pressure remains nearly constant within the combustion chamber and begins to decrease rapidly near the throat region.

A sharp pressure drop occurs at the throat as the flow transitions to sonic conditions. Downstream of the throat, the pressure continues to decrease gradually in the diverging section as the gases expand and accelerate to supersonic velocities.

The centreline pressure profile provides quantitative confirmation of the pressure reduction observed in the contour plot and demonstrates the progressive conversion of pressure energy into kinetic energy along the nozzle length.

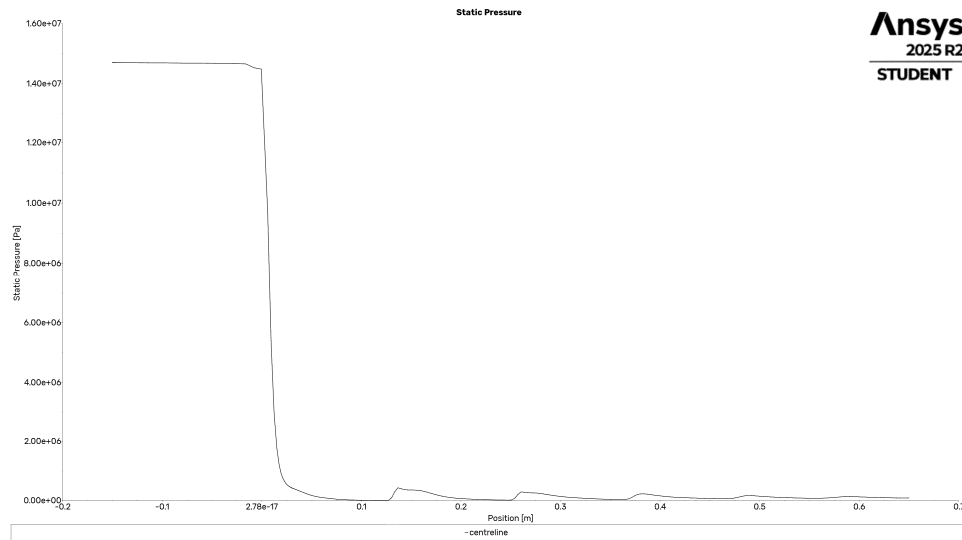


Fig. 4.11. Variation of pressure along the nozzle axis.

4.5.3 Centreline Temperature Variation

Figure. 4.12. shows the variation of static temperature along the nozzle centreline. The temperature remains relatively constant within the combustion chamber and begins to decrease near the throat region as the flow accelerates.

Downstream of the throat, the temperature continues to decrease along the diverging section as the gases expand and convert thermal energy into kinetic energy. The centreline temperature plot confirms the thermal expansion behaviour observed in the contour plot and demonstrates the progressive conversion of internal energy into exhaust kinetic energy along the nozzle length.

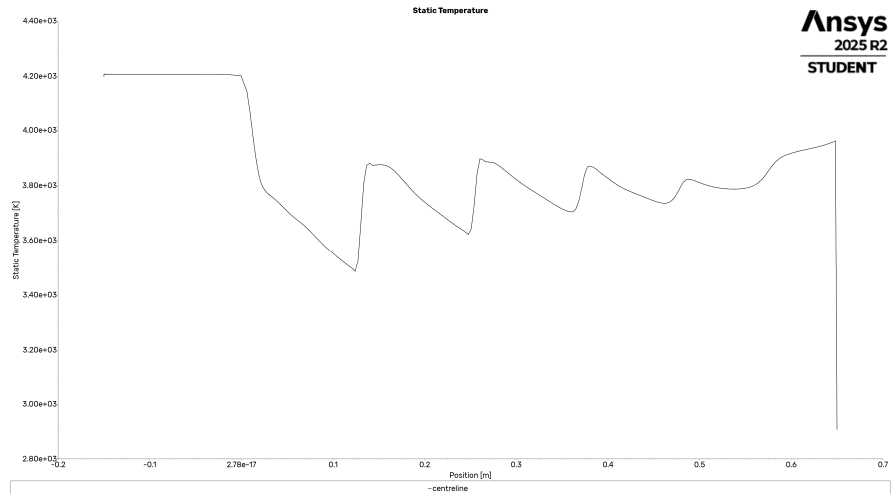


Fig. 4.12. Variation of temperature along the nozzle axis.

(With Plasma)

4.5.4 Centreline Mach Number Variation

Figure. 4.13. shows the variation of Mach number along the nozzle centreline for the plasma-assisted case. The Mach number remains below unity within the combustion chamber and increases rapidly near the throat region.

At the throat, the Mach number reaches approximately Mach 1, confirming the presence of choked flow. Downstream of the throat, the Mach number increases significantly along the diverging section as the gases expand to supersonic velocities.

The oscillations observed in the supersonic region correspond to expansion and compression waves within the exhaust jet, which are characteristic features of high-speed compressible flows in rocket nozzles.

The centreline Mach number profile provides quantitative confirmation of the subsonic–sonic–supersonic transition occurring within the nozzle during plasma-assisted combustion operation.

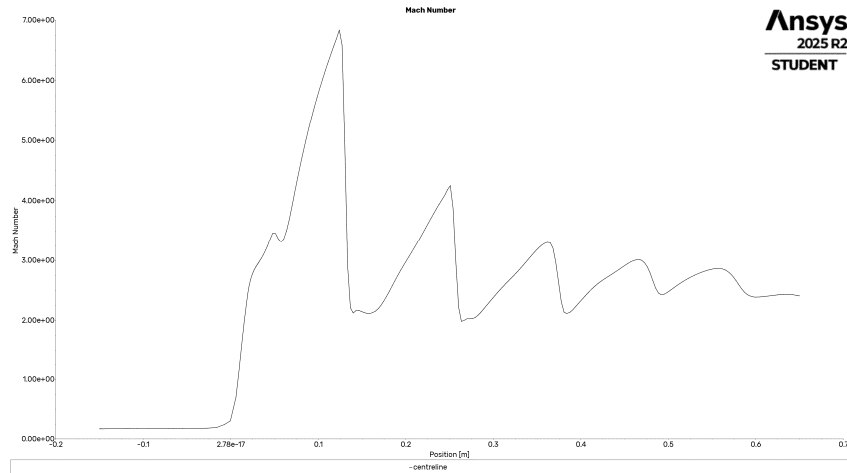


Fig. 4.13. Mach number variation along the nozzle centreline for the plasma-assisted case.

4.5.5 Centreline Pressure Variation

Figure. 4.14. presents the variation of static pressure along the nozzle centreline for the plasma-assisted case. The plot shows that the pressure remains nearly constant within the combustion chamber and decreases rapidly near the throat region.

A sharp pressure drop occurs at the throat as the flow reaches sonic conditions. Downstream of the throat, the pressure continues to decrease gradually along the diverging section as the gases expand and accelerate to supersonic velocities.

The centreline pressure profile confirms the pressure reduction observed in the contour plot and demonstrates the conversion of pressure energy into kinetic energy along the nozzle length during gas expansion.

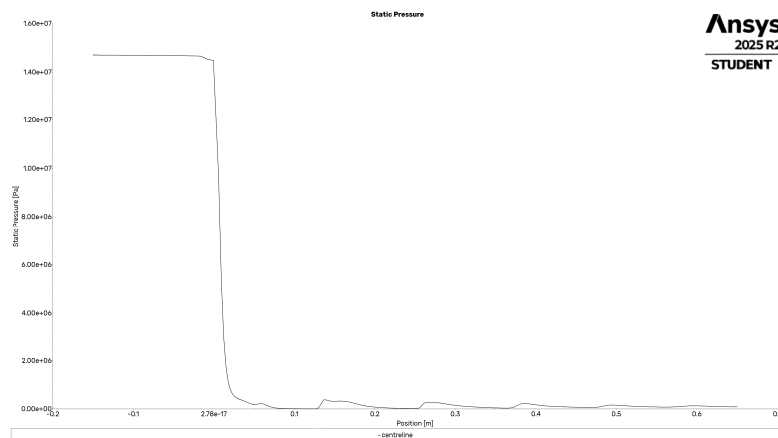


Fig. 4.14. Static pressure variation along the nozzle centreline for the plasma-assisted case.

4.5.6 Centreline Temperature Variation

Figure. 4.15. shows the variation of static temperature along the nozzle centreline for the plasma-assisted case. The temperature remains relatively constant within the combustion chamber and begins to decrease near the throat as the flow accelerates.

Downstream of the throat, the temperature continues to decrease gradually along the diverging section due to the expansion of the gases and the conversion of thermal energy into kinetic energy.

The centreline temperature profile confirms the thermal expansion behaviour observed in the contour plot and demonstrates the conversion of internal energy into kinetic energy during supersonic nozzle expansion.

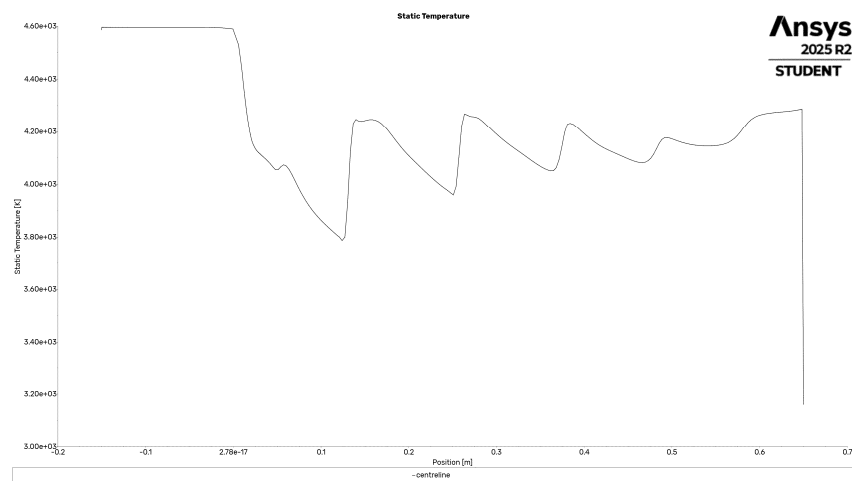


Fig. 4.15. Static temperature variation along the nozzle centreline for the plasma-assisted case.

The area-weighted average Mach number at the nozzle exit is approximately 3.39, indicating fully developed supersonic flow and efficient expansion within the converging–diverging nozzle. As shown in Fig. 4.16. the results confirm smooth flow acceleration and enhanced expansion characteristics due to improved combustion conditions.

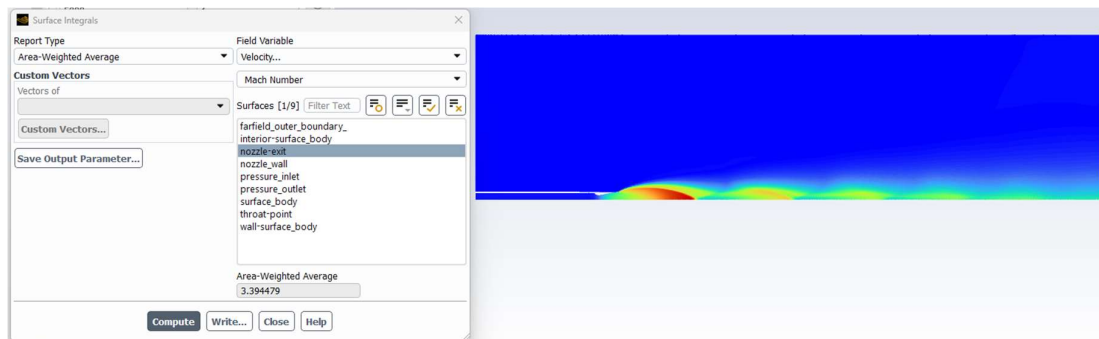


Fig. 4.16. Mach number contour at nozzle exit with plasma-assisted combustion

S.NO	Parameters	RPA (Thermochemical analysis) Baseline	Theoretical calculations	CFD Without localized plasma	CFD With localized plasma	% Change
1	Chamber Pressure (Pc)	15 MPa	15 MPa	15 MPa	15 MPa	0 %
2	Chamber Temperature (Tc)	4211 K	4211 K	4211 K	4450 K	+5.67 %
3	Exit Mach (Me) (At Nozzle Exit)	2.67	2.8	2.9	3.3	+13.79 %
4	Exit Velocity (Ve) (At Nozzle Exit)	2750 m/s	2600 m/s	2500 m/s	2900 m/s	+16.00 %
5	Exit Pressure (Pe) (At Nozzle Exit)	0.55 MPa	0.55 MPa	0.45 MPa	0.59 MPa	+31.11 %
6	Mass flow Rate (m) (At Nozzle Exit)	0.413 kg/s	0.40 kg/s	0.488 kg/s	0.511 kg/s	+4.71 %
7	Thrust (At Nozzle Exit)	1.19 kN	1.15 kN	1.26 kN	1.40 kN	+11.11 %

Table. 4.1. Comparison of Hybrid Rocket Performance Parameters with and without Plasma-Assisted Combustion

Table 4.1 summarizes the performance parameters obtained from thermochemical analysis, analytical calculations, and CFD simulations for both the baseline and plasma-assisted cases. The results indicate that the plasma-assisted case produces slightly higher exhaust velocity and thrust due to the increased combustion chamber temperature. This enhancement

demonstrates the potential of plasma-assisted combustion to improve hybrid rocket propulsion performance.

4.6 Validation with Thermochemical Analysis

To verify the accuracy of the CFD simulations, the numerical results were compared with the thermochemical predictions obtained from Rocket Propulsion Analysis (RPA). The RPA calculations predicted a combustion chamber temperature of approximately 4211 K and an exit Mach number of about 2.67 for the designed hybrid rocket nozzle.

The CFD baseline case (without plasma interaction) shows that the flow reaches sonic conditions at the nozzle throat and accelerates to supersonic velocities in the diverging section, producing an exit Mach number close to the theoretical prediction. The velocity, pressure, and temperature distributions obtained from the CFD simulation follow the expected compressible flow behaviour predicted by thermochemical analysis.

In the plasma-assisted case, the combustion chamber temperature was increased to represent the additional energy deposition generated by nanosecond pulsed plasma discharge. The simulation results show a slightly higher exhaust velocity and Mach number compared to the baseline case due to the increased thermal energy available for expansion.

4.7 Discussion of Results

The CFD simulations demonstrate that the designed hybrid rocket thrust chamber and converging–diverging nozzle are capable of producing the required compressible flow expansion for efficient propulsion. The velocity and Mach number contours clearly show the acceleration of combustion gases from subsonic conditions in the combustion chamber to supersonic velocities at the nozzle exit.

The pressure and temperature distributions obtained from the simulations follow the expected trends predicted by compressible flow theory. A gradual decrease in pressure and temperature along the nozzle length indicates the conversion of thermal and pressure energy into kinetic energy during gas expansion.

A comparison between the baseline case and the plasma-assisted case shows that increasing the combustion chamber temperature results in slightly higher exhaust velocity and Mach number. This behaviour occurs because plasma-assisted combustion increases the available

thermal energy within the combustion chamber, which enhances the expansion of the gases in the nozzle.

Although the improvement in propulsion performance is relatively moderate, the results demonstrate that plasma-assisted combustion has the potential to enhance hybrid rocket propulsion by increasing combustion efficiency and improving the exhaust flow characteristics.

4.8 Summary

This chapter presented the CFD simulation results for the designed hybrid rocket thrust chamber and converging–diverging nozzle under both baseline and plasma-assisted combustion conditions. The baseline CFD results demonstrated the expected compressible flow behaviour within the nozzle. The velocity contours showed the acceleration of combustion gases from the combustion chamber toward the nozzle exit. The pressure distribution revealed a significant reduction along the nozzle length as the gases expanded through the converging–diverging section. Similarly, the temperature contours showed a gradual decrease due to the conversion of thermal energy into kinetic energy during gas expansion. The Mach number distribution confirmed the transition from subsonic flow in the combustion chamber to sonic conditions at the throat and supersonic flow in the diverging section of the nozzle. The plasma-assisted case showed slightly higher chamber temperature, exhaust velocity, and Mach number due to the additional thermal energy introduced by plasma-enhanced combustion. The comparison between the two cases demonstrated that plasma-assisted combustion can provide a modest improvement in propulsion performance. Overall, the results confirm the effectiveness of the designed nozzle geometry and demonstrate the capability of CFD analysis to evaluate the internal flow characteristics of hybrid rocket propulsion systems with and without plasma-assisted combustion.

CHAPTER V

CONCLUSIONS AND FUTUREWORK

This chapter summarizes the major findings obtained from the design and computational analysis of the hybrid rocket thrust chamber and converging–diverging nozzle. The study focused on evaluating the compressible flow characteristics inside the propulsion system using thermochemical analysis and computational fluid dynamics (CFD) simulations.

Thermochemical calculations were performed using Rocket Propulsion Analysis (RPA) to determine combustion properties of the propellant system, while CFD simulations were conducted to analyze the flow behaviour of combustion gases within the rocket nozzle. Two simulation cases were considered: a baseline hybrid rocket configuration and a plasma-assisted combustion case, where the plasma effect was represented through an increase in combustion chamber temperature to account for additional energy deposition.

5.1 Major Conclusions

Based on the analytical calculations, thermochemical analysis, and CFD simulation results obtained in this study, the following conclusions can be drawn:

1. Hybrid Rocket Nozzle Design

A converging–diverging nozzle suitable for hybrid rocket propulsion was successfully designed using classical rocket propulsion principles. The selected geometry parameters ensured proper expansion of combustion gases from the combustion chamber to the nozzle exit and enabled the development of supersonic exhaust flow.

2. Thermochemical Analysis of Propellant System

Thermochemical calculations performed using Rocket Propulsion Analysis (RPA) provided the combustion properties required for the numerical simulations. The analysis predicted a combustion chamber temperature of approximately 4211 K, a specific heat ratio of approximately 1.22, and an exit Mach number of approximately 2.67. These parameters served as reference conditions for the CFD simulations.

3. Compressible Flow Behaviour in the Nozzle

The CFD simulation results demonstrated the expected compressible flow characteristics within the hybrid rocket nozzle. The velocity contours showed gradual acceleration of

combustion gases from the combustion chamber toward the nozzle throat, followed by rapid acceleration in the diverging section.

4. Pressure Expansion in the Nozzle

The pressure distribution showed a continuous decrease from the combustion chamber to the nozzle exit due to the expansion of high-pressure combustion gases. This pressure reduction is consistent with theoretical compressible flow relations governing converging–diverging rocket nozzles.

5. Temperature Variation Along the Nozzle

The temperature contours indicated a gradual reduction in temperature along the nozzle length as thermal energy was converted into kinetic energy during gas expansion. This trend follows the expected thermodynamic behaviour of adiabatic expansion in rocket propulsion systems.

6. Supersonic Flow Development

The Mach number distribution confirmed that the flow reached sonic conditions at the nozzle throat and accelerated to supersonic velocities in the diverging section of the nozzle. The exit Mach number obtained from the CFD simulation showed reasonable agreement with the thermochemical predictions obtained from the RPA analysis.

7. Potential Role of Plasma-Assisted Ignition

A comparison between the baseline and plasma-assisted cases indicated that increasing the combustion chamber temperature resulted in slightly higher exhaust velocity and Mach number. The plasma-assisted case showed a moderate increase in propulsion performance due to the additional thermal energy available for gas expansion within the nozzle.

8. Performance Enhancement

The comparative analysis demonstrated that the plasma-assisted case produced slightly higher exhaust velocity and thrust compared to the baseline configuration. Although the improvement in thrust was moderate, the results suggest that plasma-assisted combustion has the potential to enhance hybrid rocket propulsion performance.

5.2 Limitations of the Study

Although the present study provides valuable insights into the design and analysis of hybrid rocket propulsion systems, several limitations should be noted:

- Plasma-assisted combustion effects were represented using simplified thermal energy enhancement rather than detailed plasma kinetics Modelling.
- The CFD simulations were performed using a two-dimensional axisymmetric model to reduce computational cost.
- The combustion process inside the chamber was not modelled using detailed chemical reaction mechanisms.
- Experimental validation of the plasma-assisted hybrid rocket system was not conducted within the scope of this work.

5.3 Engineering Implications

The findings of this study provide useful insights for the design and development of hybrid rocket propulsion systems. The CFD results demonstrate that the designed converging-diverging nozzle is capable of producing the required compressible flow expansion and achieving supersonic exhaust velocities.

This confirms that classical rocket nozzle design principles combined with thermochemical analysis provide an effective approach for predicting internal flow behaviour in hybrid rocket engines.

The results also indicate that plasma-assisted ignition techniques could be a promising method for improving ignition reliability in hybrid rocket propulsion systems. Nanosecond pulsed plasma discharges generate energetic electrons and reactive radicals that can accelerate combustion reactions and reduce ignition delay.

5.4 Future Work

Future research can extend the present study in several directions:

- Experimental investigation of nanosecond plasma ignition systems integrated within hybrid rocket combustion chambers.
- Development of detailed plasma–chemical reaction models to simulate plasma-assisted combustion processes.
- Three-dimensional CFD simulations to capture complex flow structures and turbulence effects within rocket propulsion systems.
- Investigation of different electrode materials and plasma discharge configurations to improve ignition efficiency and durability.
- Optimization of hybrid rocket fuel formulations and oxidizer injection strategies to improve combustion efficiency and propulsion performance.

- Future experimental studies may integrate nanosecond plasma ignition electrodes within the combustion chamber to directly investigate plasma-assisted hybrid rocket ignition behaviour.

5.5 Final Remarks

The present study demonstrates that computational fluid dynamics combined with thermochemical analysis can provide valuable insights into the internal flow behaviour of hybrid rocket propulsion systems. The results confirm that the designed converging–diverging nozzle is capable of producing the desired compressible flow expansion and supersonic exhaust velocities required for efficient rocket propulsion.

Furthermore, the conceptual integration of plasma-assisted combustion represents a promising direction for improving ignition reliability and combustion performance in hybrid rocket engines. The outcomes of this work contribute to the understanding of hybrid rocket propulsion and provide a foundation for future research on plasma-assisted combustion technologies in aerospace propulsion systems.

The methodology adopted in this study demonstrates how analytical design methods, thermochemical analysis, and computational fluid dynamics can be effectively integrated for the preliminary design and performance evaluation of hybrid rocket propulsion systems. Such an approach provides a valuable framework for early-stage propulsion development where experimental testing may be limited.

Although the present work focuses on numerical analysis, the results provide useful insights that can support future experimental investigations of plasma-assisted hybrid rocket engines. Continued research in plasma-assisted combustion technologies may contribute to the development of more reliable ignition systems and improved combustion efficiency in next-generation aerospace propulsion systems.

REFERENCES

- [1] Iain D. Boyd (2006) A Simplified Numerical Approach for Simulating Electromagnetic Propulsion rocket-based combined-cycle engines, improving thrust at off-design conditions.
(<https://doi.org/10.1016/j.paerosci.2006.01.001>)
- [2] W. Kim & H. Do (2006) - Demonstrates that dielectric barrier discharge plasmas can stably anchor premixed flames on bluff bodies, reducing blowoff limits and enabling leaner combustion in aero-engine simulators.
(<https://doi.org/10.1016/j.paerosci.2006.01.001>)
- [3] Starikovskiy, A., & Aleksandrov, N. (2012). Plasma-assisted ignition and combustion. Progress in Energy and Combustion Science, 39, pp. 61–110.
(<https://doi.org/10.1016/j.pecs.2012.05.003>)
- [4] Starikovskiy & Nickolay Aleksandrov (2012). Plasma-assisted ignition and combustion
(<https://doi.org/10.1016/j.pecs.2012.05.003>)
- [5] Kolesnichenko et al (2014) - RF-induced drift velocity exceeding bulk plasma velocity yields significant thrust gain, with magnetic field curvature and plasma inhomogeneity critically influencing acceleration efficiency.
(<https://doi.org/10.1017/S002237781400110X>)
- [6] Ju & Sun (2014) - Plasma assisted combustion Dynamics and chemistry.
(<http://dx.doi.org/10.1016/j.pecs.2014.12.002>)
- [7] Ju, Yiguang & Sun, Wenting. (2015). Plasma assisted combustion: Progress, challenges, and opportunities. Combustion and Flame, 162(3), pp.529–532. DOI available.
(<http://dx.doi.org/10.1016/j.combustflame.2015.01.017>)
- [8] Yiguang Ju et al (2017) - Plasma Assisted Combustion.
(https://cefr.princeton.edu/sites/g/files/toruqf1071/files/documents/Yiguang%20Ju_1-5.pdf)
- [9] Kosarev et al (2017) – Incorporates sheath and boundary-layer effects into rotating electromagnetic-field thruster models, showing that optimal RF frequency and magnetic geometry yield high specific impulse with minimal wall erosion.
(<https://doi.org/10.1016/j.combustflame.2017.10.034>)
- [10] Frolov (2020) Plasma-assisted ignition in small rocket engines improves ignition reliability, reduces ignition energy, and boosts engine performance.
(<https://doi.org/10.1016/j.folorlv.2020.10.036>)

- [11] Kim, W., Do, H., Cappelli, M. A., & Mungal, M. G. (2020). Plasma-assisted flame holding in subsonic and supersonic flows. AIAA 48th Aerospace Sciences Meeting, AIAA-2010-263.
(<https://sppl.stanford.edu/wp-content/uploads/2020/09/KimFlameHolding.pdf>)
- [12] Nakagawa, I. (2020) Developed a reusable plasma torch igniter for hybrid rockets, capable of igniting wax-based fuels with gaseous oxygen and performing multiple restarts at up to 3 MPa chamber pressure.
(https://www.jes.or.jp/mag_eng/stem/Vol.80/No.2.05.html)
- [13] Karishma Chavda & Kedar Kulkarni (2021) - non-equilibrium plasmas generate active species that reduce ignition delay, stabilize lean flames, and cut emissions, though optimal plasma parameters for different fuels require further study.
(<https://www.iosrjournals.org/iosr-jap/papers/Vol13-issue4/Ser-2/E1304022635.pdf>)
- [14] Youbin Seol, Minsu Choi (2023) – Study on OH Radical Production Depending on the Pulse Characteristics in an Atmospheric-Pressure Nanosecond-Pulsed Plasma Jet
(<https://doi.org/10.3390/ma16103846>)
- [15] Wei-rui-zhang (2024) - Finds that plasma injection upstream of the combustor increases mixing and combustion efficiency in rocket-based combined-cycle engines, improving thrust at off-design conditions..
(<https://doi.org/10.1038/s41598-024-81145-z>)